

Arshia Akhtarkavan

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EDUCATION

2023 – Expected 2027 Bachelor of Science, **University of Arizona**, Tucson, AZ (GPA: 4.0/4.0)
W.A. Franke Honors College
Majors: Physics, Mathematics (Comprehensive Emphasis), & Astronomy
2022 – 2023 Diploma, **Saba International High School**, Frankfurt, Germany
2020 – 2022 High School, **The Atomic Energy Institute**, Tehran, Iran

RESEARCH AND WORK EXPERIENCE

Undergraduate Researcher in Astronomy Nov 2023 – Present

Steward Observatory, University of Arizona (Advisor: Prof. Egami)

- Development of software for forward-modeling of galaxy kinematics
- Processed incoming imaging and spectroscopic JWST data
- Conducted research on high-redshift galaxies using JWST NIRCам imaging and grism spectroscopy
- Designed and trained deep learning models to help in identifying and resolving conflicts in JWST data
- Identified and analyzed transient and variable sources in JWST survey data
- Worked full-time during summer and winter breaks and part-time during the academic year

Grader, Mathematics Department Aug 2025 – Present

Mathematics Department, University of Arizona, Tucson, AZ

- Graded weekly problem sets and quizzes for upper-division mathematics courses (MATH 454: ODEs and Stability Theory; MATH 313: Introduction to Linear Algebra)
- Wrote detailed solution manuals for homework assignments and quizzes
- Graded midterm examinations with written feedback emphasizing mathematical rigor and clarity
- Identified common conceptual errors and collaborated with instructors to address them

Research in Mathematical Physics 2025 – Present

Mathematics Department, University of Arizona (Advisor: Prof. Cherkis)

[Islands Project](#)

- Conducted independent research on Nahm equations, constructing new and higher-order solutions
- Rebuilt the Islands Project website to support student engagement in mathematical physics research
- Recipient of RII Undergraduate Research Award supporting this project

Internship in a Mechanical Engineering Workshop Jan – Mar 2023

FH Aachen University, Essen, Germany

- Gained expertise in operating basic machinery and learned practical skills in a workshop setting.
- Developed the ability to interpret and create schematics and diagrams.
- Acquired hands-on experience in using various tools and equipment for different tasks (calipers, lathes, saws, column drills, milling machines, hand tools, bench vices, etc.).

PUBLICATIONS

1. The First Photometric Evidence of a Transient/Variable Source at $z > 5$ with JWST
DeCoursey C., Egami E., Sun F., **Akhtarkavan A.**, et al. (2025; [arXiv:2504.17007](#)).

2. Slitless Areal Pure-Parallel High-Redshift Emission Survey (SAPPHIRES): Early Data Release of Deep JWST/NIRCam Images and Spectra in MACS J0416 Parallel Field
Sun F., Fudamoto Y., Lin X., Helton J. M., Hsiao T. Y-Y., Egami E., **Akhtarkavan A.**, et al. (2025; [arXiv:2503.15587](#)).
3. Discovery of 39 Transients/Variables ($z_{\text{host}} = 0.437\text{--}5.271$) in GOODS-N using JADES and CONGRESS JWST/NIRCam Images
DeCoursey C., Egami E., Sun F., **Akhtarkavan A.**, et al. (2024; [Transient Name Server AstroNote](#)).

CONFERENCE PRESENTATIONS

1. **Forward-Modeling Emission Line Galaxy Kinematics and Morphology with JWST NIR-Cam Grisms**
iposter presentation, 247th Meeting of the American Astronomical Society (AAS 247), January 2026
2. **Forward-Modeling Emission-Line Galaxy Kinematics and Morphology with JWST NIR-Cam Grisms**
Contributed Talk, Steward Observatory Internal Symposium, University of Arizona, March 2026
3. **Islands project and the Nahm Equations**
Talk and Seminar, University of Arizona, April 2026

AWARDS AND SCHOLARSHIPS

Dorrit Hoffleit Research Scholarship, Yale University (2026)
 College of Science Galileo Circle Scholar (Mathematics), University of Arizona (2026)
 College of Science Galileo Circle Scholar (Physics), University of Arizona (2026)
 Weaver Award for Undergraduate Research (2026)
 Kenneth S. Krane Physics Scholarship (2025)
 Evelyn O. Bychinsky Promising Astronomer Award (2025)
 RII-Sponsored Campuswide Undergraduate Student-Initiated Original Research (2025)
 Glenn C. Purviance Scholarship in Physics (2023)
 Academic Year Highest Academic Distinction (2023 – Present)
 Dean’s List with Distinction - Full Time (2023 – Present)
 Global Wildcat Tuition Award, University of Arizona (2023 – Present)
 Silver Medal, National Astronomy and Astrophysics Olympiad (2022)

SKILLS

Programming Languages	Python, C/C++, SQL
Scientific & Data Analysis Libraries (Python)	NumPy, Matplotlib, PyTorch, Astropy, SciPy, Specutils, Photutils, Pandas, JAX
Astronomy & Astrophysics Software	DS9, JWST Pipeline, Source detection and Photometry, Webbpsf, NIRCam imaging and grism spectroscopy
Software & Research Systems	Linux/Unix, Git/GitHub, HPC environments, remote computing (ssh), GPU-accelerated computing (CUDA), Jupyter, VS Code, Emacs, LaTeX, Microsoft Office
Concepts	Data Processing Pipelines, Scientific Computing, Machine Learning, Numerical Methods

LANGUAGES

English	Fluent
Farsi (Persian)	Native
German	B1 level
Mandarin Chinese	Currently learning